

# TITAN-X

## Executive Business Plan · 2026–2030

Stratospheric persistent aerial platform for venture capitalists and business owners — regional infrastructure between aircraft and satellites.



Firefighting configuration — retardant and water delivery. California is the planning early concentration.

## 1. Executive summary

**TITAN-X** is a free-flight stratospheric aerostat with a **20,000 lb** modular payload, indefinite endurance on helium buoyancy and solar power, and mission reconfiguration by swapping gondolas and underbelly modules.

**Investment thesis:** fund pathfinder → first full-scale hull → California firefighting + metro nodes → planning **12-unit** fleet by 2030. Pre-positioning converts response from days to hours; civil and commercial utilization underwrites year-round availability; dual-use configs share one cost base.

|                                            |                                             |                                         |
|--------------------------------------------|---------------------------------------------|-----------------------------------------|
| <b>\$13M</b><br>Unit CAPEX (fleet volume)  | <b>\$156M</b><br>12-unit fleet CAPEX        | <b>\$2.4M</b><br>Annual OPEX / unit     |
| <b>\$200K</b><br>Monthly break-even / unit | <b>17.8 : 1</b><br>Planning 10-yr fleet ROI | <b>20,000 lb</b><br>Net modular payload |

## 2. Platform at a glance

| Parameter                                                                                                                                                                                                                      | Value            | Notes                          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|--------------------------------|
| Size                                                                                                                                                                                                                           | 480 ft × 200 ft  | Swept manta planform           |
| Net payload                                                                                                                                                                                                                    | 20,000 lbs       | ≈ school-bus weight            |
| Altitude range                                                                                                                                                                                                                 | 500 – 130,000 ft | Tree-top to lower stratosphere |
| Endurance                                                                                                                                                                                                                      | Indefinite       | Solar + buoyancy               |
| Ops carbon (persistent)                                                                                                                                                                                                        | Near-zero        | No combustion for loiter       |
| Architecture                                                                                                                                                                                                                   | Fully modular    | Gondola / underbelly swap      |
| <ul style="list-style-type: none"> <li>• Rockoon loft to ~100,000 ft for propellant savings, then recover the platform</li> <li>• Civil, commercial, and defense requirements share the same platform and cost base</li> </ul> |                  |                                |



Formation / persistent high-altitude operations — regional pre-positioning on one modular hull family.

### 3. Problem & solution

Wildfire and disaster response still depends on scarce surge aircraft. Satellites persist but cannot deliver mass. Cities need hours-scale airborne nodes for observation, temporary communications, and logistics.

**TITAN-X fills the band between aircraft and satellites:** indefinite loiter, school-bus-class modular payload, near-zero persistent ops carbon, and pre-positioned regional coverage instead of surge-only response.

### 4. Regional deployment

#### Major cities & high-value regions

One TITAN-X (or a pair) per region, loaded with modular response packages, on station in hours — disaster response, communications restoration, observation, logistics.

#### California firefighting concentration

- Fire-behavior, thermal, and smoke-plume sensors
- Retardant or water delivery modules
- Incident command and communications packages
- Decision-support for incident commanders

Modules shift to other missions in lower-risk periods to preserve utilization.

#### Smaller complementary platforms

Secondary observation, relay, weather, and lower-cost coverage under full-size regional nodes.

### 5. Customers & missions

| Segment                               | Use                                              |
|---------------------------------------|--------------------------------------------------|
| State / regional wildfire authorities | CA firefighting concentration                    |
| Metro & high-value regions            | Persistent disaster & comms nodes                |
| Critical infrastructure               | Monitoring and logistics support                 |
| Maritime / shipping                   | Domain awareness and ship support                |
| Public safety                         | Airborne threat awareness (low-cost drones)      |
| Science / weather                     | High-altitude observation                        |
| Rockoon / launch customers            | Stratospheric loft + recover                     |
| Dual-use / defense (selected)         | ISR, relay, forward logistics, responsive access |

## 6. Unit economics

| Item                                    | Planning value        |
|-----------------------------------------|-----------------------|
| Unit CAPEX (fleet volume)               | \$13.0M               |
| 12-unit fleet CAPEX                     | \$156M                |
| Annual OPEX per unit                    | \$2.4M                |
| Break-even utilization                  | ~\$200K / month       |
| Mature utilization target (base)        | \$8–12M / unit / year |
| Contribution at \$10M rev / \$2.4M opex | ~\$7.6M / unit / year |

### Illustrative mature revenue mix

| Stream                                    | Share  |
|-------------------------------------------|--------|
| Regional standby / availability           | 35–45% |
| Wildfire / disaster mission fees          | 20–30% |
| Infrastructure / maritime / public safety | 15–25% |
| Rockoon / science / specialty             | 5–15%  |
| Dual-use / defense options                | 0–20%  |

## 7. Financial plan 2026–2030

USD millions. Anchored to consolidated summary CAPEX / OPEX / break-even. The planning 17.8 : 1 ten-year fleet ROI is a long-horizon target on the stood-up fleet — not a 2030 cash guarantee.

### Fleet build (base)

|                         | 2026 | 2027 | 2028 | 2029 | 2030 |
|-------------------------|------|------|------|------|------|
| Full-size units EOY     | 0*   | 1    | 3    | 6    | 12   |
| CA firefighting-primary | —    | 1    | 2    | 3    | 4    |
| Metro / regional nodes  | —    | 0    | 1    | 3    | 8    |

\*Pathfinder year — endurance, station-keeping, modular swap validation.

### Capital expenditure (base, \$M)

|                                  | 2026 | 2027 | 2028 | 2029 | 2030 | Total |
|----------------------------------|------|------|------|------|------|-------|
| Pathfinder + engineering         | 18   | 6    | 4    | 3    | 2    | 33    |
| Hull production (learning curve) | —    | 18   | 30   | 42   | 78   | 168   |
| Ground / ops infrastructure      | 4    | 6    | 8    | 10   | 12   | 40    |
| Total CAPEX                      | 22   | 30   | 42   | 55   | 92   | 241   |

### Operating model (base, \$M)

|  | 2026 | 2027 | 2028 | 2029 | 2030 |
|--|------|------|------|------|------|
|--|------|------|------|------|------|

|                      |       |       |     |     |     |
|----------------------|-------|-------|-----|-----|-----|
| Avg units in service | 0     | 0.5   | 2.0 | 4.5 | 9.0 |
| Revenue              | 0.5   | 4     | 18  | 42  | 90  |
| Direct OPEX          | 1.5   | 3.5   | 7   | 13  | 24  |
| Program / G&A        | 4     | 5     | 7   | 9   | 11  |
| EBITDA (planning)    | (5.0) | (4.5) | 4   | 20  | 55  |

2030 check: ~\$90M revenue on ~9 average units ≈ **\$10M per unit-year**, above the \$2.4M OPEX break-even.

## Scenario fan — 2030

| Case         | Units | Revenue | EBITDA |
|--------------|-------|---------|--------|
| Conservative | 6     | \$45M   | \$18M  |
| Base         | 12    | \$90M   | \$55M  |
| Upside       | 12+   | \$130M  | \$85M  |

## 8. Use of funds & milestones

| Use                                   | Share  |
|---------------------------------------|--------|
| Pathfinder + first full-scale hull    | 35–45% |
| CA firefighting + first metro modules | 20–30% |
| Production tooling / learning-curve   | 15–20% |
| Ops bases, helium, insurance, crew    | 10–15% |
| Corporate / BD / compliance           | 5–10%  |

Gate heavy fleet CAPEX on endurance demo, modular swap demo, and first paid availability or mission contract.

## 9. Risks

- Airworthiness / airspace — pathfinder before fleet claims
- Early unit cost above \$13M volume target — learning-curve modeled explicitly
- Public contracting cycles — civil commercial utilization underwrites standby
- 17.8 : 1 ROI is a planning ten-year fleet figure, not a 2026–2030 guarantee

## 10. The ask

We are engaging venture capitalists and business owners to fund or co-develop:

- Pathfinder validation and first full-scale hull
- California firefighting concentration as the first revenue-bearing mission stack
- Metro pre-positioning nodes that turn TITAN-X into infrastructure

**Bottom line.** Indefinite endurance. 20,000 lb modular payload. Near-zero persistent ops carbon. Pre-positioned regional coverage with a California firefighting emphasis. Dual-use economics on one hull family. **2026–2030 builds the twelve-unit base; utilization after that is where the summary ROI lives.**